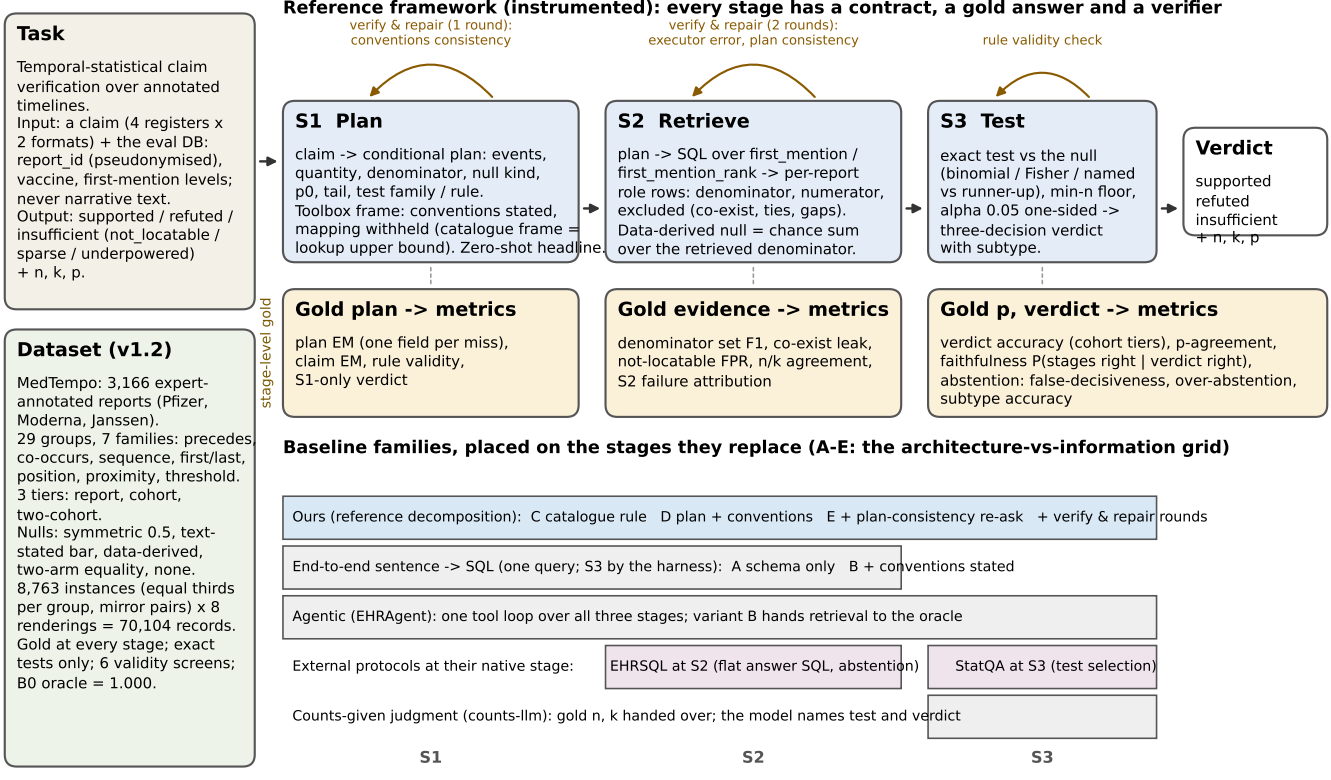
Reference framework (instrumented): every stage has a contract, a gold answer and a verifier
verify & repair (1 round): conventions consistency
verify & repair (2 rounds): executor error, plan consistency
rule validity check
Task
Temporal-statistical claim verification over annotated timelines.
Input: a claim (4 registers x 2 formats) + the eval DB: report_id (pseudonymised), vaccine, first-mention levels; never narrative text.
Output: supported / refuted / insufficient (not_locatable / sparse / underpowered)
+ n, k, p.
Dataset (v1.2)
MedTempo: 3,166 expert-annotated reports (Pfizer, Moderna, Janssen).
29 groups, 7 families: precedes, co-occurs, sequence, first/last, position, proximity, threshold.
3 tiers: report, cohort, two-cohort.
Nulls: symmetric 0.5, text-stated bar, data-derived, two-arm equality, none.
8,763 instances (equal thirds per group, mirror pairs) x 8 renderings = 70,104 records.
Gold at every stage; exact tests only; 6 validity screens; B0 oracle = 1.000.
S1 Plan
claim -> conditional plan: events, quantity, denominator, null kind, p0, tail, test family / rule.
Toolbox frame: conventions stated, mapping withheld (catalogue frame = lookup upper bound). Zero-shot headline.
S2 Retrieve
plan -> SQL over first_mention / first_mention_rank -> per-report role rows: denominator, numerator, excluded (co-exist, ties, gaps).
Data-derived null = chance sum over the retrieved denominator.
S3 Test
exact test vs the null (binomial / Fisher / named vs runner-up), min-n floor, alpha 0.05 one-sided -> three-decision verdict with subtype.
Verdict
supported
refuted
insufficient
+ n, k, p
stage-level gold
Gold plan -> metrics
plan EM (one field per miss), claim EM, rule validity, S1-only verdict
Gold evidence -> metrics
denominator set F1, co-exist leak, not-locatable FPR, n/k agreement, S2 failure attribution
Gold p, verdict -> metrics
verdict accuracy (cohort tiers), p-agreement, faithfulness P(stages right | verdict right), abstention: false-decisiveness, over-abstention, subtype accuracy
Baseline families, placed on the stages they replace (A-E: the architecture-vs-information grid)
Ours (reference decomposition): C catalogue rule D plan + conventions E + plan-consistency re-ask + verify & repair rounds
End-to-end sentence -> SQL (one query; S3 by the harness): A schema only B + conventions stated
Agentic (EHRAgent): one tool loop over all three stages; variant B hands retrieval to the oracle
External protocols at their native stage:
EHRSQL at S2 (flat answer SQL, abstention)
StatQA at S3 (test selection)
Counts-given judgment (counts-llm): gold n, k handed over; the model names test and verdict
S1
S2
S3